Lathika N 241801135 exp 4

```
!pip install nltk -q

import pandas as pd
import numpy as np
import re
import string
import nltk

from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity

nltk.download('punkt')
nltk.download('punkt_tab')
nltk.download('stopwords')

df = pd.read_csv("/content/Reviews.csv")

print("Shape:", df.shape)
display(df.head())
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt.zip.
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt_tab.zip.
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Unzipping corpora/stopwords.zip.
Shape: (106337, 10)
```

	Id	ProductId	UserId	ProfileName	HelpfulnessNumerator	He
0	1	B001E4KFG0	A3SGXH7AUHU8GW	delmartian	1.0	
1	2	B00813GRG4	A1D87F6ZCVE5NK	dll pa	0.0	
2	3	B000LQOCH0	ABXLMWJIXXAIN	Natalia Corres "Natalia Corres"	1.0	
3	4	B000UA0QIQ	A395BORC6FGVXV	Karl	3.0	
4	5	B006K2ZZ7K	A1UQRSCLF8GW1T	Michael D. Bigham "M. Wassir"	0.0	

```
df = df[['Text']].dropna()
df = df.drop_duplicates()
df = df.head(10000).reset_index(drop=True)

print("Number of reviews:", len(df))
display(df.head())
```

```
Number of reviews: 10000
```

	Text
0	I have bought several of the Vitality canned d...
1	Product arrived labeled as Jumbo Salted Peanut...
2	This is a confection that has been around a fe...
3	If you are looking for the secret ingredient i...
4	Great taffy at a great price. There was a wid...

```
stop_words = set(stopwords.words('english'))

def preprocess(text):
    text = text.lower()
    text = re.sub(r'[^a-zA-Z\s]', '', text)
    tokens = word_tokenize(text)
    tokens = [word for word in tokens if word not in stop_words]
    return ' '.join(tokens)

df['Cleaned_Text'] = df['Text'].apply(preprocess)

display(df[['Text', 'Cleaned_Text']].head())
```

	Text	Cleaned_Text
0	I have bought several of the Vitality canned d...	bought several vitality canned dog food produc...
1	Product arrived labeled as Jumbo Salted Peanut...	product arrived labeled jumbo salted peanutsth...
2	This is a confection that has been around a fe...	confection around centuries light pillowy citr...
3	If you are looking for the secret ingredient i...	looking secret ingredient robitussin believe f...
4	Great taffy at a great price. There was a wid...	great taffy great price wide assortment yummy ...

```
vectorizer = TfidfVectorizer()

tfidf_matrix = vectorizer.fit_transform(df['Cleaned_Text'])

print("TF-IDF Matrix Shape:", tfidf_matrix.shape)
print("Vocabulary Size:", len(vectorizer.vocabulary_))
```

```
TF-IDF Matrix Shape: (10000, 24267)
Vocabulary Size: 24267
```

```
def retrieve_reviews(query):
    query_cleaned = preprocess(query)
```

```
    query_vector = vectorizer.transform([query_cleaned])

    similarity = cosine_similarity(query_vector, tfidf_matrix).flatten()

    top_indices = similarity.argsort()[-5:][::-1]

    results = df.iloc[top_indices].copy()
    results['Similarity Score'] = similarity[top_indices]

    return results[['Similarity Score', 'Text', 'Cleaned_Text']]
```

```
queries = [
    "great product with fast shipping",
    "disappointed",
    "excellent taste",
    "poor packaging"
]

for query in queries:
    print("\n" + "="*80)
    print("QUERY:", query)
    print("="*80)
    display(retrieve_reviews(query))
```

```
for query in queries:
    results = retrieve_reviews(query)

    print("\nQuery:", query)
    print("Average Similarity Score:",
          round(results['Similarity Score'].mean(), 4))

    print("Top Review:")
    print(results.iloc[0]['Text'][:300])
    print()
```

```
Query: great product with fast shipping
Average Similarity Score: 0.4246
Top Review:
Enjoyed the product and they also provided very fast shipping. I'm about out
```

6011	0.401147	The energy drink is a great product. The shipp...	energy drink great product shipping price craz...
7048	0.392853	My daughter lives in Hawaii and sent me some g...	daughter lives hawaii sent great coffee keurig...